

LLM, SLM, or Hybrid Utilization: Decision Factors Across Performance, Efficiency, and Domain – A Systematic Literature Review

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Abstract

Large language models (LLMs) deliver strong reasoning and generation performance, but their cost, latency, energy demand, and hardware requirements remain high. Small language models (SLMs) are better suited to edge and privacy-sensitive deployment, yet their capabilities vary across domains. This paper presents a systematic literature review of 54 studies on collaborative SLM–LLM systems spanning hybrid, multi-agent, retrieval-centered, and ensemble architectures, with monolithic LLMs treated as baselines rather than as a primary architecture class. Guided by four research questions on performance, efficiency, orchestration, and domain suitability, we apply a Kitchenham-guided review protocol to synthesize models, architectures, datasets, metrics, latency, cost, energy, and reported limitations. The literature shows a clear shift toward hybrid and multi-agent designs, but cost and energy remain far less reported than accuracy. Overall, SLMs are most competitive when they are paired with routing, retrieval, planning, or fusion mechanisms in narrow, structured, or domain-specific settings.

Keywords

systematic literature review, large language models, small language models, hybrid architectures, multi-agent systems, edge-cloud inference

1. Introduction

Large language models (LLMs) excel at reasoning, coding, and multimodal tasks [2, 3, 4], but their hardware and deployment requirements often confine them to cloud infrastructure or high-end local systems, raising privacy, cloud dependence, and cost concerns [5, 6, 7, 8].

Small language models (SLMs) are attractive for local deployment and lower-cost, data-sovereign operation. Fine-tuning or distillation can also make them competitive on domain-specific tasks [9, 10, 11]. They pair naturally with routing, verification, and planning mechanisms [8, 6, 12]. However, SLMs still trail frontier LLMs on broad reasoning and tasks requiring extensive contextual knowledge [4, 13], which has driven recent work on hybrid and agent-based architectures that combine small and large models for a better capability–efficiency trade-off [14, 15, 16].

This is primarily a systems-design question. Across domains, hybrid and multi-agent architectures aim to improve not only accuracy but also latency, cost, and computational efficiency through edge–cloud verification, expert routing, retrieval, and planner-based orchestration [14, 17, 18, 12, 19, 20, 5]. These developments motivate a closer examination of how collaborative architectures allocate work between small and large models, when monolithic LLM baselines remain necessary, and when orchestration provides the best balance between capability and efficiency.

Accordingly, this review examines when hybrid or multi-agent SLM–LLM architectures provide a better trade-off between capability and efficiency than monolithic LLM baselines, and under which domain conditions routing, retrieval, planning, and role specialization allow smaller models to contribute

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effectively.

The main contributions are threefold: (1) a synthesis of 54 studies on hybrid SLM–LLM architectures, retrieval-centered collaboration, ensembles, and multi-agent workflows; (2) a review structure organized around four practical questions on performance, efficiency, orchestration, and domain suitability; and (3) an analytical matrix mapping each study to the research question(s) it addresses.

2. Related Works

Prior review literature on SLMs and SLM–LLM collaboration includes four closely related works. The closest is a systematic literature review (SLR) guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement, with a reproducible multi-database protocol. The study adopts a broad scope on SLMs and offers limited coverage of the 2025–2026 wave of hybrid and agent-based systems [21]. Three narrative surveys cover SLM definitions, training techniques, collaboration patterns, trustworthiness, and edge deployment. They do not follow a formal literature review framework [22, 23, 24]. While existing surveys comprehensively detail SLM training techniques and model-level reasoning capabilities, they largely treat collaboration as a conceptual capability. Our study addresses this gap through a Kitchenham-guided synthesis that maps the actual architectural infrastructure—specifically, how routing mechanisms, edge–cloud bandwidth, and task decomposition directly affect operational constraints such as latency, API cost, and energy consumption.

3. Research Methodology

This section outlines the review protocol, which follows the systematic literature review guidelines of Kitchenham and Charters [1]. It covers the research questions, search strategy, eligibility criteria, study selection, and data extraction and synthesis process. PRISMA 2020 is used as a reporting aid for documenting the study-selection flow rather than as the review methodology itself.

3.1. Research Questions

To systematically evaluate the technical and operational trade-offs between collaborative SLM–LLM systems and monolithic LLM baselines, the review protocol is structured around the following four core research questions (RQs):

- **RQ1 (Performance):** How does the performance (accuracy/efficacy) of multi-agent SLMs or hybrid architectures (LLM + SLM) compare with that of monolithic LLM baselines in specific reasoning and classification tasks?
- **RQ2 (Efficiency):** What is the impact of hybrid and multi-agent architectures on computational efficiency, specifically regarding inference cost, latency, and energy consumption, compared to monolithic LLM baselines or full-LLM inference?
- **RQ3 (Orchestration):** What orchestration and routing mechanisms are employed to manage the division of labor between large and small models in collaborative systems?
- **RQ4 (Domain Suitability):** In which specific domains (e.g., healthcare, coding, telecommunications, and edge or multimodal settings) do collaborative SLM-centered architectures achieve performance comparable to monolithic LLM baselines, and where do they fail?

3.2. Information Sources and Search Strategy

To capture a comprehensive and high-quality corpus of contemporary collaborative language model architectures, the literature search was centered on the Web of Science (WoS) Core Collection, with the final search performed on 8 February 2026. Scopus and the Association for Computing Machinery (ACM) Digital Library were also checked with the same core concepts as coverage validation; this check did not identify additional eligible studies that would materially alter the final corpus.

The search strategy was systematically developed by extracting the core concepts from the defined research questions and refining the terminology through initial pilot searches in the artificial intelligence (AI) literature. This iterative process helped identify common synonyms and alternate phrasing used in the literature (e.g., “Compact Model” for SLMs, or “Agentic” for multi-agent systems). To guarantee that every retrieved study examined the intersection of model size, collaborative architectural design, and system evaluation, the final query was structured into four mandatory components and applied to the title, abstract, and keyword fields: (“*Large Language Model**” OR “*LLM*” OR “*Generative AI*” OR “*Foundation Model**”) AND (“*Small Language Model**” OR “*SLM*” OR “*Compact Model**” OR “*Lightweight Model**”) AND (“*Multi-agent*” OR “*Hybrid*” OR “*Orchestration*” OR “*Routing*” OR “*Ensemble*” OR “*Agentic*”) AND (“*Performance*” OR “*Accuracy*” OR “*Efficiency*” OR “*Latency*” OR “*Cost*” OR “*Benchmark*”).

Because the review targets collaborative SLM–LLM systems, all four components were mandatory in the search string. Monolithic LLMs were considered only as baselines reported inside the included studies, not as standalone primary studies.

3.3. Eligibility Criteria

To systematically isolate systems-level engineering research from the broader generative AI literature, retrieved records were evaluated against strict eligibility criteria. A comprehensive breakdown of the inclusion and exclusion parameters applied during the screening phases is presented in Table 1.

Table 1
Inclusion and Exclusion Criteria for Study Selection

Criterion	Include	Exclude
Architecture Scope	Collaborative SLM–LLM systems using routing, retrieval, verification, fallback, or role specialization.	SLM-only papers or monolithic LLM comparisons without a collaborative SLM component; monolithic LLMs were allowed only as baselines inside otherwise eligible studies.
Model Size and Transparency	Identifiable SLM component with reported or inferable size not exceeding 7B parameters; closed-box LLMs allowed only as baseline, teacher, or fallback components.	Primary architecture depends on closed-box application programming interface (API)-bound models with undisclosed details, or no reproducible SLM component can be identified.
Topic Focus	Reports system-level metrics such as performance, latency, memory, cost, or orchestration efficiency.	Focuses only on ethical, sociological, or legal issues without technical benchmarks.
Data Quality	Contains empirical results and experimental validation using benchmarks or custom datasets.	Purely theoretical papers or non-empirical narrative surveys.

3.4. Study Selection Process

Following the Kitchenham and Charters review protocol [1], candidate studies were screened against the predefined eligibility criteria and assessed at title, abstract, and full-text stages. The selection flow is reported using PRISMA 2020 terminology [25]. The initial search yielded 116 records. During identification and screening, the title, abstract, and keywords of all records were reviewed manually, excluding 36 records that failed the primary eligibility criteria, including non-technical narrative surveys, extended abstracts without full empirical validation, and studies in unrelated domains. The remaining 80 reports advanced to full-text assessment, where 26 additional records were excluded due to model size or architecture mismatch (n=18), topic or scope mismatch (n=5), or lack of original empirical data or a remaining narrative-survey character (n=3). To improve objectivity, inclusion and exclusion decisions at both stages were made by majority vote, with ties resolved through collaborative discussion and escalation to the academic advisors, resulting in a final corpus of 54 primary studies.

3.5. Data Extraction and Synthesis Strategy

To systematically address the research questions, a standardized data extraction protocol was applied to the final corpus of 54 primary studies. For each study, data was extracted into a structured matrix to

capture both the functional design of the systems and the empirical evidence used to justify them.

The extracted fields included the specific SLM and LLM models utilized; the overarching architecture type (e.g., Hybrid, Multi-Agent, Retrieval-Augmented Generation (RAG), Ensemble); orchestration mechanisms; evaluation metrics; baseline models; and quantitative performance indicators (accuracy, latency, cost, and energy). To ensure analytical rigor, each extracted variable was explicitly mapped to one or more of the core Research Questions (RQs). This mapping, together with concise field definitions, is detailed in Table 2.

Table 2

RQ × data extraction matrix. Check marks indicate direct support; Arch. = architecture, Orchestr. = orchestration, and Base. = baseline.

Field	Description	RQ1	RQ2	RQ3	RQ4
SLM(s)	Specific small language model(s) evaluated, fine-tuned, or deployed.	✓			✓
LLM(s)	Large language model(s) used as a baseline, teacher, or fallback.	✓			✓
Arch. Type	Overall system design (e.g., hybrid, multi-agent, RAG, ensemble).	✓			✓
Orchestr. Method	Coordination mechanism such as routing, decomposition, fusion, or voting.			✓	✓
Agent Count	Number of distinct agents, experts, or role-specific components.		✓	✓	✓
Task Type	Downstream application, domain, or problem setting evaluated.	✓		✓	✓
Dataset	Benchmark, open-source corpus, or proprietary data used in evaluation.	✓		✓	
Metrics	Evaluation criteria used to assess performance and efficiency.	✓	✓	✓	✓
LLM Base.	Standalone performance score of the comparison large model.	✓		✓	✓
SLM Base.	Standalone performance score of the comparison small model.	✓		✓	✓
Accuracy	Final performance score achieved by the proposed architecture.	✓		✓	
Latency (ms)	Reported inference delay, throughput, or time-to-response.		✓	✓	✓
Cost (\$)	Reported API cost or computational savings versus full LLM use.		✓	✓	✓
Energy	Power use, joules-per-token, or carbon-related efficiency metric.		✓	✓	✓
Findings	Main study takeaway on SLM–LLM collaboration or orchestration.	✓	✓	✓	✓
Limits	Reported constraints, failure modes, or deployment bottlenecks.		✓	✓	

The full comparative analysis table and the comprehensive data extraction spreadsheet are available at <https://bit.ly/SLM-LLM-Data-2026>.

Primary studies mapping: S1: [2], S2: [3], S3: [4], S4: [5], S5: [6], S6: [7], S7: [8], S8: [9], S9: [10], S10: [11], S11: [12], S12: [13], S13: [14], S14: [15], S15: [16], S16: [17], S17: [18], S18: [19], S19: [20], S20: [26], S21: [27], S22: [28], S23: [29], S24: [30], S25: [31], S26: [32], S27: [33], S28: [34], S29: [35], S30: [36], S31: [37], S32: [38], S33: [39], S34: [40], S35: [41], S36: [42], S37: [43], S38: [44], S39: [45], S40: [46], S41: [47], S42: [48], S43: [49], S44: [50], S45: [51], S46: [52], S47: [53], S48: [54], S49: [55], S50: [56], S51: [57], S52: [58], S53: [59], and S54: [60].

Following data extraction, a narrative synthesis approach was utilized. Because the included studies operate across diverse domains (e.g., telecommunications, healthcare, software engineering) and report varying metrics, a purely statistical meta-analysis was not feasible. Instead, studies were grouped by their architectural taxonomy, allowing for a comparative analysis of how different orchestration

methods and domain constraints impact the performance and efficiency trade-offs between large and small models.

Data Provenance and Synthesis. To formalize the narrative synthesis and ensure evidence traceability, a thematic coding methodology was applied to the extracted data. Studies were grouped not by parameter size, but by their dominant architectural mechanism (e.g., routing, retrieval, or multi-agent debate). Quantitative metrics, including accuracy, latency, and cost, were extracted exactly as reported. Where numerical metrics were absent—such as for the latency overhead in S1—limitations and operational constraints were either extracted verbatim or inferred strictly from conclusions explicitly stated by the primary authors, thereby avoiding post hoc reviewer assumptions. This structured thematic grouping forms the basis of the architectural concept matrix in Table 3.

4. Findings

The included studies cluster into four recurring architectural patterns. Table 3 summarizes these patterns by linking their dominant orchestration mechanisms to the engineering constraints they address and the primary studies assigned to each category. This concept matrix provides the organizing structure for the research-question-based analysis that follows.

Table 3
SLM–LLM architectural concept matrix.

Architectural Pattern	Orchestration Mechanism	Key Engineering Constraints Addressed	Primary Studies
Hybrid Architectures	Token-level routing, draft-verification, uncertainty-aware skipping, and edge–cloud distribution.	API cost, edge–cloud bandwidth, and latency thresholds.	S2, S13, S27, S32, S33, S45, S46, S48, S49, S53.
Multi-Agent Systems	Role specialization, iterative verification, adversarial debate, and feedback loops.	Reasoning depth, logical accuracy, and complex task decomposition.	S1, S6, S7, S11, S15, S16, S20, S29, S37, S43, S51, S52, S54.
Retrieval-Centered (RAG)	Adaptive retrieval planning, context expansion, fact-grounding, and action-space constraint.	Hallucination reduction, data privacy, and strict domain accuracy.	S4, S10, S21, S30, S31, S44.
Ensembles and Heterogeneous Fusion	Tabular transformation, cross-modal late fusion, and prompt-guided multimodal sampling.	Hardware limits on central processing units (CPUs) and edge devices, and non-text modality processing.	S3, S5, S8, S9, S12, S14, S17, S18, S19, S22, S23, S24, S25, S26, S28, S34, S35, S36, S38, S39, S40, S41, S42, S47, S50.

4.1. Systematic Map of the Literature

Before addressing the specific research questions, we present a systematic map of the 54 included primary studies to provide visual context for the field’s current trajectory. As illustrated in Figure 1, the distribution of publication years reveals a sharp acceleration in SLM–LLM collaborative research. While early foundational work on model adaptation appeared in 2023, the vast majority of the corpus (39 studies, 72.2%) was published in 2025, underscoring that architectural orchestration—rather than standalone model scaling—is a rapidly maturing engineering priority.

Figure 2 visualizes the distribution of applied domains across the literature. The studies are heavily concentrated in areas requiring strict logical reasoning, such as software engineering, mathematics, and related logical-reasoning tasks (11 studies, 20.4%), and fields governed by severe privacy and latency constraints, namely healthcare and medicine (10 studies, 18.5%) and telecommunications, edge, and networking (10 studies, 18.5%). This distribution confirms that hybrid and multi-agent systems are

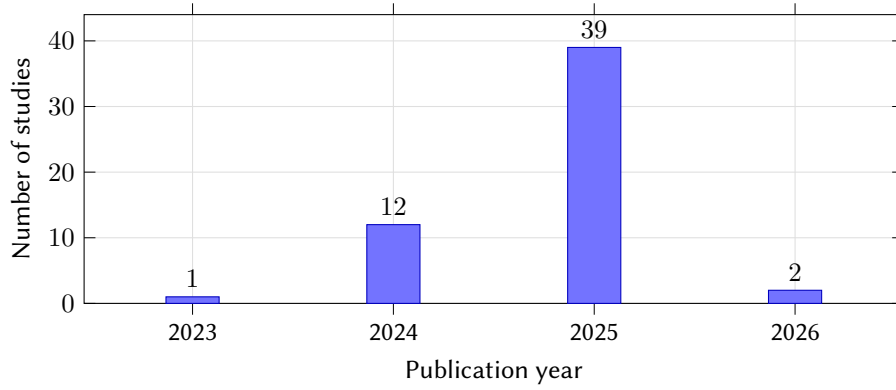


Figure 1: Publication-year distribution of the 54 included primary studies.

primarily engineered to solve domain-specific operational bottlenecks rather than general-purpose generative tasks.

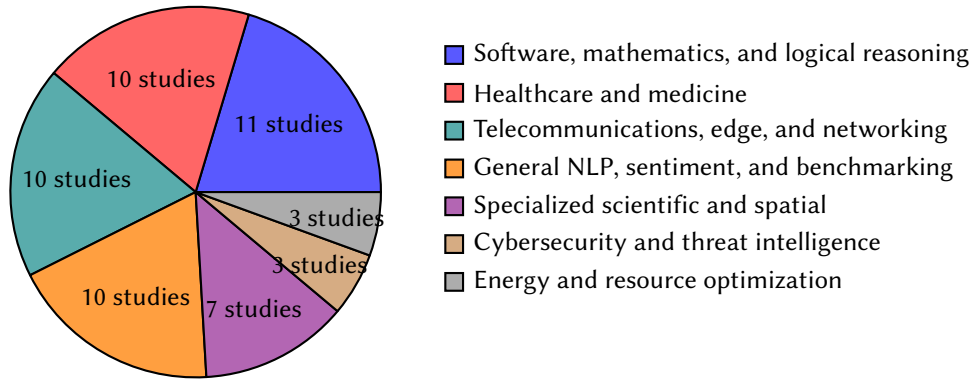


Figure 2: Primary applied-domain distribution of the 54 included studies. Categories reflect each study’s principal evaluation domain; methodologically overlapping studies are assigned to one primary category for this aggregate view.

4.2. RQ1: Comparative Performance of Hybrid and Multi-Agent Systems

The review shows that both hybrid and multi-agent architectures can outperform standalone small-model baselines, but their advantages emerge through different forms of coordination rather than parameter scaling alone. Hybrid systems achieve their strongest gains when computation is selectively distributed between local small models and stronger remote models. Token-level edge–cloud collaboration improves small-model performance without requiring full-query escalation: draft-verification, low-confidence routing, and uncertainty-aware skipping allow small models to approach LLM-level quality while invoking the larger model selectively (S13, S48, S49). At the sub-task level, structured task decomposition and cloud–edge allocation also maintain reasoning accuracy comparable to the best baselines (S14). In domain-specific settings, hybrid orchestration can outperform larger general-purpose models: a fine-tuned Phi-2 (2.7B) combined with retrieval, reranking, hybrid search, and long-context support achieves 84.20% accuracy, exceeding GPT-4o mini (82.05%) and GPT-4o (80.80%) on telecommunications question answering (QA) (S17).

Multi-agent systems, by contrast, show their comparative strength when performance depends on role specialization, critique, verification, and distributed reasoning. A three-agent Phi-3-based architecture performs on par with human experts in conceptual genomics tasks, although its gains weaken on more complex code-generation workflows, indicating that multi-agent benefits remain task-sensitive (S52). In reasoning-intensive question answering, a Proponent–Opponent–Arbitrator structure built on

compact models such as Qwen2.5 0.5B and DeepSeek-R1 1.5B surpasses both single-agent reasoning and vanilla LLM baselines (S43). Likewise, assistant-checker collaboration supported by memory and feedback loops enables small-model multi-agent systems to approach or exceed larger baselines across database, knowledge-graph, and code-correction tasks (S29). Overall, the findings indicate that hybrid systems are most effective when the main challenge is balancing accuracy with latency and cost through adaptive routing, whereas multi-agent systems are most effective when the task benefits from explicit decomposition of reasoning roles and iterative verification. Thus, the comparative performance advantage of both architecture families is driven primarily by orchestration strategy, not by model size alone.

Key takeaways

- Hybrid systems are strongest when selective escalation can balance model quality against latency and cost.
- Multi-agent systems benefit tasks that require role specialization, critique, and iterative verification, but their gains remain task-sensitive.
- Orchestration strategy explains comparative performance more consistently than model size alone.

4.3. RQ2: Efficiency, Cost, Latency, and Energy Trade-offs

Efficiency motivates many hybrid and multi-agent systems, but the impact on cost, latency, and energy depends heavily on orchestration. Hybridization is not a universal efficiency solution; it often shifts bottlenecks rather than eliminating them.

Regarding inference cost and token consumption, architectures employing selective escalation or task decomposition drastically reduce reliance on massive standalone LLMs. S49 achieves an over 80% cost reduction compared to full LLM inference by routing fewer than 7% of generated tokens to the cloud. Similarly, S13 uses a token-level draft-verification method to match the quality of a 70B parameter LLM while consuming only 25.8% to 31.2% of standard cloud inference costs. Query-level decomposition yields similar savings: S14 cuts average API costs by 83.57% via parallel reasoning. Computational savings are also heavily emphasized through token reduction. S40 achieves a 95% reduction in LLM token transmissions via federated uncertainty-aware routing, and S15 eliminates redundant multi-round communications through an SLM-generated directed acyclic graph, reducing token consumption by 35% to 70%. In dialogue state tracking, S16 highlights that a retrieval-based router reduces overall floating-point operations (FLOPs) by approximately 62%.

Conversely, the impact on latency is highly polarized. In optimal scenarios, relying on lightweight models accelerates throughput: a fine-tuned SLM in a hybrid classification pipeline (S10) operates at 7.41 ms on a graphics processing unit (GPU), and uncertainty-aware skipping in an edge-cloud setup provides $2.54\times$ faster token throughput (S48). However, orchestration mechanisms frequently introduce new bottlenecks. Planners and retrievers add temporal overhead, as shown by the adaptive retrieval planner in S11 (1.42 seconds versus 0.95 seconds for naive retrieval) and by cross-encoder reranking in S17. Multi-agent collaboration also multiplies inference cycles: S43 reports a $1.8\times$ inference-time overhead for adversarial debate, while complex agentic workflows can push average response latency to 107 seconds (S31). Processing time is further affected by hardware and modality constraints, including constrained local hardware, dual-path video sampling, CPU-only ensembles, and key-value cache recomputation at the edge (S20, S23, S50, S5).

Energy consumption remains the least documented metric in the reviewed studies, and most reported efficiency gains are resource proxies rather than direct energy measurements. The clearest energy-centered evidence comes from S4, which argues for energy-per-token as a more informative metric and shows that forcing a small model to execute complex reasoning workflows, such as Chain-of-Thought, can substantially increase its per-query energy use. In some settings, a larger model prompted more directly is therefore more energy-efficient than a smaller model that generates a long reasoning trace. By contrast, S29 reports a 26.4% reduction in computational redundancy through dual-scale memory,

which may indicate lower resource demand but does not by itself constitute a direct energy benchmark. Taken together, the review suggests that energy efficiency depends on orchestration and generation length as much as on model size, and that stronger claims require standardized, hardware-specific measurement protocols.

Key takeaways

- Selective routing and task decomposition can substantially reduce API use, token transmission, and computational demand.
- Latency gains are conditional: routing can accelerate inference, while retrieval, planning, and multi-agent loops can introduce significant overhead.
- Energy claims remain weak without direct, hardware-specific measurements that account for generation length and orchestration cost.

4.4. RQ3: Orchestration and Routing Mechanisms

The studies contain a broad set of orchestration mechanisms, but they can be organized into a small number of recurring families. Across the included studies, performance differences are explained less by absolute model size than by how effectively the system retrieves evidence, routes uncertainty, and divides labor across components.

Token-level orchestration is most visible in verification-oriented hybrid systems. S13 uses the SLM as a drafter and the LLM as a verifier on difficult tokens. S48 extends this pattern with uncertainty-aware skipping so that likely acceptable SLM tokens can bypass the remote model. S49 reaches a related design point by routing only low-confidence tokens for external inference. Taken together, these studies show that token-level collaboration provides fine-grained control over answer quality, but it can also introduce communication and synchronization overhead when escalation rates increase.

Query-level routing appears when the system selects a processing path for the full input rather than for individual tokens. S16 exemplifies retrieval-based expert routing at the dialogue level, while S10 shows that instance-level model selection can also be effective in customer-review analysis. These designs are most useful when “easy” and “hard” cases can be separated in a semantically meaningful way before full generation begins.

RAG is a major control mechanism across the included studies even when the architecture is classified as hybrid rather than purely RAG. S17, S11, S39, and S47 use retrieval not only to expand context but also to constrain the action space of a smaller model. In these studies, retrieval acts as a coordination layer that narrows ambiguity, grounds the answer in domain evidence, and reduces reliance on raw model scale alone.

Task decomposition is increasingly important in the recent literature. S14 uses a task decomposer, dependency graph, and plug-and-play adapter to allocate subtasks between local and remote models. S53 uses a feedback-integrated workflow that gathers structured feedback from multiple perspectives before revising the prompt of the smaller model. Related role-based decomposition also appears in S23, S29, and S52. Across these studies, the common idea is that small models remain viable when the workflow can be partitioned into specialist stages rather than handled as a single undifferentiated generation step.

Orchestration frequently extends beyond unimodal text processing to integrate diverse data modalities and specialized neural architectures. Several frameworks are designed to fuse language models with non-text representations and structured data. For example, S20 combines coarse-grained and fine-grained visual semantics through prompt-guided video sampling, while S8 fuses language-model embeddings with graph-convolutional features for molecular-property prediction. This heterogeneous routing is particularly prevalent in applied domains such as medicine; S19 orchestrates convolutional networks (ConvNeXt) for anatomical localization alongside a self-reflective LLM agent, and S34 employs a late-fusion approach to combine structured gradient boosting models based on electronic health record (EHR) tabular data with unstructured clinical text analysis. In these settings, orchestration operates as a structured fusion mechanism, ensuring that visual, spatial, or mathematical data is processed by the optimal specialized architecture before language generation occurs.

Key takeaways

- Token- and query-level routing provide different levels of control over when a larger model is invoked.
- Retrieval acts as a control mechanism by grounding responses and constraining the smaller model's action space.
- Task decomposition and heterogeneous fusion enable specialist components to collaborate, but add coordination and synchronization costs.

4.5. RQ4: Domain Suitability and Application-Specific Constraints

The choice between hybrid and multi-agent systems is rarely domain-agnostic. Privacy mandates, latency budgets, and verification demands largely determine the orchestration method. Across the 54 primary studies, distinct architectural preferences emerge across four core applied domains.

In fields with strict privacy constraints or high-stakes accuracy requirements, hybridization is driven by the need to keep data local while grounding the SLM in verifiable facts. Local SLMs fine-tuned on clinical corpora and augmented by RAG are therefore well suited to medical reporting and clinical classification tasks (S19, S30), while few-shot open-source models can be competitive for structured radiology labeling (S12). Related healthcare studies also show that local ensembles, tabular transformations, and distillation can support diagnostic or summarization workflows without relying on a fully remote LLM (S28, S33, S41). In scientific settings, structurally augmented SLMs can be effective when language representations are fused with domain-specific models, as in molecular property prediction with graph-convolutional features (S8).

Tasks requiring rigid syntax, iterative refinement, and logical debugging benefit from multi-agent or feedback-driven SLMs rather than zero-shot massive LLMs. Pedagogy, assessment, code correction, and database tasks favor workflows in which small models receive structure, memory, or explicit feedback (S6, S29, S32). In software and bioinformatics settings, role-specific agents can mimic developer-reviewer workflows and generate or revise code offline, although broad static analysis still favors larger models when the task requires open-ended type inference (S24, S2, S52, S53).

Telecommunications architectures prioritize latency budgets, bandwidth allocation, and domain-specific vernacular, making hybrid edge–cloud systems and retrieval-grounded SLMs the dominant pattern. Telecom QA benefits from fine-tuning, retrieval, reranking, and long-context support because these mechanisms ground smaller generators in technical standards and network terminology (S17, S47). Beyond QA, latency optimization and bandwidth reduction drive hybridization through lightweight prediction, uncertainty-aware skipping, and sub-task decomposition (S37, S48, S14).

In edge, cyber-physical, and multimodal settings, orchestration is governed by hardware limits and input modality. Mission-critical aerospace and geospatial workflows rely on local or distributed agents to process operational data under deployment constraints (S23, S45). Industrial video monitoring and cross-modal sensing benefit from heterogeneous pipelines in which local models handle coarse or sensor-side processing while larger multimodal models are invoked only for complex cases (S20, S1). Overall, these domains show that collaborative SLM–LLM systems are strongest when the workflow can be partitioned according to privacy, latency, modality, or verification needs.

Key takeaways

- Healthcare and other privacy-sensitive domains favor local, retrieval-grounded models with verifiable evidence.
- Software, reasoning, and telecommunications tasks benefit from structured feedback, specialist agents, or latency-aware edge–cloud routing.
- Domain constraints determine architectural suitability; no single hybrid or multi-agent design is consistently best across applications.

4.6. Cross-RQ Synthesis

Viewed together, the four research questions suggest a consistent systems principle: hybrid language-model design works best when the large model is treated as a selectively invoked capability rather than the default engine for every token, query, or subtask.

Strong results are associated with explicit control layers such as routers, planners, retrievers, or fusion modules. Future progress is therefore unlikely to come only from making SLMs slightly larger or LLMs slightly cheaper; it will also depend on better confidence estimation, task decomposition, retrieval selection, and cross-model scheduling.

Methodologically, because accuracy dominates reporting, many studies overstate practical efficiency. The review includes cases where orchestration clearly reduces cost or latency, but also cases where coordination merely shifts the bottleneck. A mature evaluation protocol should therefore report resource metrics as first-class outcomes. Application-wise, the strongest SLM-centered deployments are domain-adapted systems in which the small model is grounded, specialized, or protected by escalation logic. This is why telecom, structured reasoning, radiology labeling, optimisation formulation, and industrial monitoring are prominent in the positive findings.

4.7. Implications for Practitioners

For software engineers and system architects, the main design trade-offs are operational rather than purely model-centric. In practice, latency, cost, privacy, and hardware constraints often dominate deployment decisions. Four practical design patterns emerge from the reviewed studies. When strict data privacy is the main constraint, local SLM + RAG architectures are most suitable for healthcare, telecom, and defense settings (S17, S23, S19, S30), but their performance depends heavily on retriever quality. When high throughput is required, hybrid edge–cloud architectures are appropriate for Internet of Things (IoT) and network traffic scenarios (S37, S40, S48, S49), reducing API costs while requiring careful threshold tuning. For logical reasoning tasks such as code and mathematics, multi-agent SLM designs can improve accuracy (S29, S15, S43, S52), but they introduce significant latency overhead. For multimodal data, heterogeneous fusion is better suited to medical imaging and robotics (S8, S20, S19, S1), although the slowest model in the pipeline often becomes the bottleneck. Three practitioner-facing principles follow: (1) **Design for routing**: orchestration, not model size, drives performance; (2) **Balance latency-cost trade-offs**: real-time tasks favor token skipping, while batch tasks tolerate multi-agent loops; and (3) **Treat RAG as control**: retrieval constrains the SLM action space and grounds outputs in verifiable facts.

5. Threats to Validity

This review has several threats to validity that should be stated explicitly.

Internal validity. Although the review follows a Kitchenham-guided protocol and uses PRISMA terminology to report the selection flow, study identification, screening, and extraction still involve researcher judgment, especially for borderline papers in which the SLM component or empirical contribution was not equally central. To reduce this risk, borderline screening decisions were discussed through team consensus, and the synthesis remained tied to the predefined research questions and the same extraction matrix throughout.

Construct validity. Core constructs such as “SLM”, “hybrid architecture”, “multi-agent system”, and “efficiency” are not standardized across the literature. A single hybrid label may refer to token-level verification, retrieval-grounded generation, planner-based decomposition, or multimodal fusion, while efficiency may mean latency, FLOPs, API cost, bandwidth reduction, or energy use. This limits direct comparability because studies grouped under the same high-level category may differ substantially in mechanism. To partially limit this, the review separates broad architecture labels from extracted orchestration mechanisms and discusses recurring mechanism families explicitly.

Conclusion validity. Cross-study inference is limited by the uneven and inconsistent reporting of evaluation outcomes. Accuracy is reported in most papers, but latency, cost, and energy appear only in a minority, often with different units, hardware assumptions, or deployment settings, which prevents formal meta-analysis and limits claims about efficiency gains across the included studies. Publication and reporting bias may further favor positive hybrid results, so the extraction also recorded reported limitations, and the synthesis emphasizes uneven reporting and task-sensitive trade-offs rather than universal gains.

External validity. The study set is based on a Web of Science-centered search and a bounded review window, which improves transparency but may miss relevant studies from other databases, preprint platforms, or fast-moving venues. Although WoS covers many major peer-reviewed journals and conference proceedings, the field evolves quickly and the current corpus is heavily concentrated in 2025 and in a limited set of application domains. Therefore, the conclusions should be generalized cautiously beyond the covered time period, tasks, and deployment settings.

6. Conclusion

This paper presented a systematic literature review of 54 studies on collaborative SLM–LLM systems across hybrid, multi-agent, ensemble, and retrieval-centered architectures, with monolithic LLMs treated as baselines rather than as a primary study class. Across the four review questions, a consistent pattern emerges: SLMs are most competitive when they are supported by retrieval, routing, uncertainty estimation, task decomposition, prompt optimisation, or multimodal fusion rather than used as isolated replacements for larger models. The literature is stronger in architectural innovation than in evaluation discipline. Many studies report accuracy improvements, but far fewer report latency, cost, or energy in a form that supports fair comparison. The current evidence is therefore sufficient to show that hybridization can work, but often insufficient to show when it is the better deployment choice under realistic resource constraints. The strongest positive results remain concentrated in narrower or more structured settings such as telecom question answering, selected healthcare tasks, optimisation, and industrial monitoring. The energy evidence is especially limited: direct energy measurement is rare, and the clearest energy-centered result shows that when small models are pushed into long Chain-of-Thought traces, their per-query energy use can exceed that of a larger model prompted more directly [5]. Other studies report only proxy improvements, such as reduced computational redundancy [35], which do not establish verified energy savings. Future work should therefore treat energy as a first-class evaluation outcome and report energy-per-token, reasoning length, hardware configuration, and escalation behavior under comparable conditions.

A. Online Appendix

The online appendix contains the full comparative analysis table for the 54 primary studies and the comprehensive data extraction spreadsheet used in this review.

Access link: <https://bit.ly/SLM-LLM-Data-2026>

Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Declaration on Generative AI

During the preparation of this work, the authors used OpenAI Codex for formatting assistance and for limited paraphrasing and rewording related to the CEUR-ART template migration and compliance

wording. The authors reviewed and edited the resulting content and take full responsibility for the publication's content.

References

- [1] Kitchenham, B., Charters, S.: Guidelines for Performing Systematic Literature Reviews in Software Engineering. EBSE Technical Report EBSE-2007-01, Keele University and Durham University (2007).
- [2] Esteban-Romero, S., Martín-Fernández, I., Gil-Martín, M., Fernández-Martínez, F.: Synthesizing Olfactory Understanding: Multimodal Language Models for Image-Text Smell Matching. *Symmetry* **17**(8), 1349 (2025). doi:10.3390/sym17081349
- [3] Shivarpatna Venkatesh, A.P., Sunil, R., Sabu, S., Reis, M.S., Bodden, E., Amir, M.: An Empirical Study of Large Language Models for Type and Call Graph Analysis in Python and JavaScript. arXiv preprint arXiv:2410.00603 (2025). doi:10.48550/arXiv.2410.00603
- [4] Chen, H., Deng, W., Yang, S., Xu, J., Jiang, Z., Ngai, E.C.H., et al.: Toward Edge General Intelligence via Large Language Models: Opportunities and Challenges. arXiv preprint arXiv:2410.18125 (2025). doi:10.48550/arXiv.2410.18125
- [5] Wilhelm, P., Wittkopp, T., Kao, O.: Beyond Test-Time Compute Strategies: Advocating Energy-per-Token in LLM Inference. In: Proceedings of the 5th Workshop on Machine Learning and Systems (EuroMLSys 2025), pp. 208–215. ACM, Rotterdam, The Netherlands (2025). doi:10.1145/3721146.3721953
- [6] Bao, R., Xue, N., Sun, Y., Chen, Z.: Dynamic Quality-Latency Aware Routing for LLM Inference in Wireless Edge-Device Networks. arXiv preprint arXiv:2508.11291 (2025). doi:10.48550/arXiv.2508.11291
- [7] Tigani, S.: CPU-Only Self Enhancing Authoring Copilot Design-Based Markov Decision Processes Orchestration and Qwen 3 Local Large Language Model. *Technologies* **13**(11), 520 (2025). doi:10.3390/technologies13110520
- [8] Aydın, S.K., Ali, R.H., Faiz, S., Khan, T.A.: An Integrated AI Framework for Personalized Nutrition Using Machine Learning and Natural Language Processing for Dietary Recommendations. *Appl. Sci.* **15**(17), 9283 (2025). doi:10.3390/app15179283
- [9] Xie, L., Jin, Y., Xu, L., Chang, S., Xu, X.: Fusing Domain Knowledge with a Fine-Tuned Large Language Model for Enhanced Molecular Property Prediction. *J. Chem. Theory Comput.* **21**(14), 6743–6758 (2025). doi:10.1021/acs.jctc.5c00605
- [10] Zhu, X., Li, J., Liu, Y., Ma, C., Wang, W.: Distilling Mathematical Reasoning Capabilities into Small Language Models. arXiv preprint arXiv:2401.11864 (2024). doi:10.48550/arXiv.2401.11864
- [11] Alabbasi, N., Alhussein, O., Muhaidat, S., Debbah, M.: Do you actually need an LLM? Rethinking language models for customer reviews analysis. arXiv preprint arXiv:2411.02555 (2024). doi:10.21203/rs.3.rs-5006214/v1
- [12] Ji, X., Chen, S., Shon, Y., et al.: RAP-RAG: A Retrieval-Augmented Generation Framework with Adaptive Retrieval Task Planning. *Electronics* **14**(1), 185 (2025). 10.3390/electronics14214269
- [13] Dorfner, F.J., Gadewar, M.R., et al.: Comparing Commercial and Open-Source Large Language Models for Labeling Chest Radiograph Reports. *Radiology* **313**(1), e240590 (2024). doi:10.1148/radiol.241139
- [14] Hao, Z., Wei, J., Wang, T., Ren, J.: Hybrid SLM and LLM for Edge-Cloud Collaborative Inference. In: Proceedings of the 2024 Workshop on Edge and Mobile Foundation Models (EdgeFM 2024), pp. 36–41. ACM, Tokyo, Japan (2024). doi:10.1145/3662006.3662067
- [15] Shao, C., Hu, X., Lin, Y., Xu, F.: Division-of-Thoughts: Harnessing Hybrid Language Model Synergy for Efficient On-Device Agents. In: Proceedings of the ACM Web Conference 2025 (WWW 2025), pp. 1822–1833. ACM, Sydney, Australia (2025). doi:10.1145/3696410.3714765
- [16] Zeng, J., Liao, J., Sun, S., Zhang, G., Xu, C., Liu, H., et al.: Multi-Agent Collaborative Reasoning via Cloud-Edge Framework. *IEEE Network* (2026). doi:10.1109/MNET.2026.3651036

- [17] Lee, C.H., Cheng, H., Ostendorf, M.: OrchestraLLM: Efficient Orchestration of Language Models for Dialogue State Tracking. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pp. 1434–1445. Association for Computational Linguistics, Mexico City, Mexico (2024). doi:10.18653/v1/2024.naacl-long.79
- [18] Alabbasi, N., Erak, O., Alhussein, O., Lotfi, I., Muhaidat, S., Debbah, M.: TeleOracle: Fine-Tuned Retrieval-Augmented Generation with Long-Context Support for Networks. arXiv preprint arXiv:2411.02617 (2024). doi:10.48550/arXiv.2411.02617
- [19] Chatzistefanidis, I., Nikaein, N.: Symbiotic Agents: A Novel Paradigm for Trustworthy AGI-driven Networks. *Comput. Networks* **273**, 111749 (2025). doi:10.1016/j.comnet.2025.111749
- [20] Han, J., Kim, M., Kim, Y., Lee, W.H.: Self-Reflective Chest X-Ray Report Generation with Clinical-Aware Detection and Multilevel Readability. *J. Med. Syst.* **49**, 190 (2025). doi:10.1007/s10916-025-02326-5
- [21] Corradini, F., Leonesi, M., Piangerelli, M.: State of the Art and Future Directions of Small Language Models: A Systematic Review. *Big Data Cogn. Comput.* **9**(7), 189 (2025). doi:10.3390/bdcc9070189
- [22] Wang, F., et al.: A Comprehensive Survey of Small Language Models in the Era of Large Language Models: Techniques, Enhancements, Applications, Collaboration with LLMs, and Trustworthiness. arXiv preprint arXiv:2411.03350 (2024). doi:10.48550/arXiv.2411.03350
- [23] Chen, Y., et al.: A Survey on Collaborative Mechanisms Between Large and Small Language Models. arXiv preprint arXiv:2505.07460 (2025). doi:10.48550/arXiv.2505.07460
- [24] Wang, L., et al.: A Survey on Collaborating Small and Large Language Models for Performance, Cost-effectiveness, Cloud-edge Privacy, and Trustworthiness. arXiv preprint arXiv:2510.13890 (2025). doi:10.48550/arXiv.2510.13890
- [25] Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., et al.: The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. *BMJ* **372**, n71 (2021). doi:10.1136/bmj.n71
- [26] Cao, Q., et al.: Toward Energy-Safe Industrial Monitoring: A Hybrid LLM Framework for Video Captioning. *Appl. Sci.* **15**(23), 12848 (2025). doi:10.3390/app152312848
- [27] Han, X., Sun, J., Yu, S., Qin, L., Luo, H., Li, Y.: CommentAgent: a LLM-Powered Agent Framework for Automated Comment Generation and Opinion Understanding. *J. Supercomput.* **81**(15), 1438 (2025). doi:10.1007/s11227-025-07957-6
- [28] Demir, E., Bilgin, M.: DPATransLLM: Detection of Pronominal Anaphora in Turkish Sentences Using Transformer-Based, Large Language Models and Hybrid Ensemble Approach. *Appl. Sci.* **15**(23), 12480 (2025). doi:10.3390/app152312480
- [29] Pacheco De Almeida Prado, J., Zammit Mangion, D., Zammit, B., Gauci, J., Muscat, A., Mizzi, S., et al.: Enhancing Flight Deck Decision Support with Distributed GenAI: A Multi-Agent Approach. In: Proceedings of the 2025 IEEE Aerospace Conference (AERO), pp. 1–20. IEEE, Big Sky, MT, USA (2025). doi:10.1109/AERO63441.2025.11068478
- [30] Zhu, J., Wang, L., Guc, Y., Wang, F.: OFSLMs: Offline Fine-Tuned Small Language Models Based on Hybrid Synthetic Knowledge for Robot Introspective Decision-Making. In: Proceedings of the 2024 IEEE International Conference on Robotics and Biomimetics (ROBIO), pp. 1839–1844. IEEE, Bangkok, Thailand (2024). doi:10.1109/ROBIO64047.2024.10907647
- [31] Jeong, S., Baek, J., Cho, S., Hwang, S., Park, J.: Test-Time Self-Adaptive Small Language Models for Question Answering. In: Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 15459–15469. Association for Computational Linguistics, Singapore (2023). doi:10.18653/v1/2023.findings-emnlp.1033
- [32] Weiss, Y., Buttery, P., Africa, D.D., Diehl Martinez, R.: Investigating ReLoRA: Effects on the Learning Dynamics of Small Language Models. arXiv preprint arXiv:2509.12960 (2025). doi:10.48550/arXiv.2509.12960
- [33] Pietrzak, Z., Mączka, K., Niemiec, M.: What if, Behind the Curtain, There Is Only an LLM? A Holistic Evaluation of TinyLlama-Generated Synthetic Cyber Threat Intelligence. *Electronics* **14**(24), 4971 (2025). doi:10.3390/electronics14244971

- [34] Gao, Y., Myers, S., Chen, S., Dligach, D., Miller, T., Bitterman, D.S., et al.: Uncertainty Estimation in Diagnosis Generation from Large Language Models: Next-Word Probability Is Not Pre-Test Probability. *JAMIA Open* **8**(1), ooae154 (2025). doi:10.1093/jamiaopen/ooae154
- [35] Xiong, Y., Huang, M., Liang, X., Tao, M.: ResilioMate: A Resilient Multi-Agent Task Executing Framework for Enhancing Small Language Models. *IEEE Access* **13**, 86892–86911 (2025). doi:10.1109/ACCESS.2025.3567244
- [36] Khader, K.A., Hussein, M.S., Abu-Issa, A.S.: Adapting Large Language Models for Arabic: Comparative Evaluation, Fine-Tuning, and Ethical Deployment. *IEEE Access* **13**, 182621–182632 (2025). doi:10.1109/ACCESS.2025.3623796
- [37] Kuligin, L., Lammert, J., Ostapenko, A., Bressemer, K., Boeker, M., Tschochoei, M.: Prompt Design for Medical Question Answering with Large Language Models. *Mach. Learn. Appl.* **22**, 100758 (2025). doi:10.1016/j.mlwa.2025.100758
- [38] Abbas, N., Atwell, E.: Cognitive Computing with Large Language Models for Student Assessment Feedback. *Big Data Cogn. Comput.* **9**(5), 112 (2025). doi:10.3390/bdcc9050112
- [39] Wang, Z., Li, K., Peng, S., Liu, L., Yang, X., Yao, K., et al.: A Weighted Voting Approach for Traditional Chinese Medicine Formula Classification Using Large Language Models: Algorithm Development and Validation Study. *JMIR Med. Inform.* **13**, e69286 (2025). doi:10.2196/69286
- [40] Dodson, R.J.B., Dodson, C.R.S., Dodson, P.S., et al.: Explainable Machine Learning-Based Prediction of Psoriatic Arthritis Flares Using Heterogenous Real-World Data for Personalised Patient Care. *Methods* **246**, 95–106 (2026). doi:10.1016/j.ymeth.2025.10.010
- [41] Deng, J., Wu, Y., Yesha, Y., Nguyen, P.: Improving VTE Identification through Language Models from Radiology Reports: A Comparative Study of Mamba, Phi-3 Mini, and BERT. *arXiv preprint arXiv:2408.09043* (2024). doi:10.48550/arXiv.2408.09043
- [42] Das, S., Srihari, R.: Compos Mentis at SemEval2024 Task6: A Multi-Faceted Role-Based Large Language Model Ensemble to Detect Hallucination. In: *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pp. 1449–1454. Association for Computational Linguistics, Mexico City, Mexico (2024). doi:10.18653/v1/2024.semeval-1.208
- [43] Xu, X., Zhao, J., Zhang, Y., Li, R.: Integrating Reinforcement Learning and LLM with Self-Optimization Network System. *Network* **5**(3), 39 (2025). doi:10.3390/network5030039
- [44] Erak, O., Alabbasi, N., Alhussein, O., Lotfi, I., Hussein, A., Muhaidat, S., et al.: Leveraging Fine-Tuned Retrieval-Augmented Generation with Long-Context Support: For 3GPP Standards. *arXiv preprint arXiv:2408.11775* (2025). doi:10.48550/arXiv.2408.11775
- [45] Afzal, A., Vladika, J., Fazlija, G., Staradubets, A., Matthes, F.: Towards Optimizing a Retrieval Augmented Generation using Large Language Model on Academic Data. In: *Proceedings of the 2024 8th International Conference on Natural Language Processing and Information Retrieval (NLPIR 2024)*, pp. 250–257. ACM, Okayama, Japan (2025). doi:10.1145/3711542.3711575
- [46] Solat, F., Lee, J., Seif, M., Niyato, D., Poor, H.V.: Federated Learning-Enabled Hybrid Language Models for Communication-Efficient Token Transmission. *IEEE Internet Things J.* **12**(24), 53574–53592 (2025). doi:10.1109/JIOT.2025.3618095
- [47] Lee, C., Kumar, S., Vogt, K.A., Munshi, M., Tallapudi, P., Vogt, A., et al.: Democratizing Cost-Effective, Agentic Artificial Intelligence to Multilingual Medical Summarization Through Knowledge Distillation. *Sci. Rep.* **15**, 27619 (2025). doi:10.1038/s41598-025-10451-x
- [48] Jin, X., Wang, Z., Chen, J., Yang, L., Oh, B., Hwang, S., et al.: HLMEA: Unsupervised Entity Alignment Based on Hybrid Language Models. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 11, pp. 11888–11896. AAAI Press, Philadelphia, PA, USA (2025). doi:10.1609/aaai.v39i11.33294
- [49] Huynh, T., Nguyen, D.: Collaborating Agents with Large Language Models for Question-Answering Tasks. In: *Proceedings of the 5th IEEE International Conference on Software Engineering and Artificial Intelligence (SEAI)*, pp. 1–5. IEEE, Fuzhou, China (2025). doi:10.1109/SEAI65851.2025.11108177
- [50] Al Daoud, E., Al Daoud, L., Asassfeh, M., Al-Shaikh, A., Al-Sherideh, A.S., Afaneh, S.: Enhancing Cybersecurity with Transformers: Preventing Phishing Emails and Social Media Scams. In: *Proceedings of the 2024 IEEE Conference on Dependable and Secure Computing (DSC)*, pp. 31–36.

IEEE, Tokyo, Japan (2024). doi:10.1109/DSC63325.2024.00017

- [51] Ying, S., Li, Z., Yu, M.: Beyond Words: Evaluating Large Language Models in Transportation Planning. arXiv preprint arXiv:2409.14516 (2024). doi:10.48550/arXiv.2409.14516
- [52] Pintelas, E., Koursaris, A., Livieris, I.E., Tampakas, V.: TextNeX: Text Network of eXperts for Robust Text Classification-Case Study on Machine-Generated-Text Detection. *Mathematics* **13**(10), 1555 (2025). doi:10.3390/math13101555
- [53] Gichamba, A., Idris, T.K., Ebiyau, B., Nyberg, E., Mitamura, T.: ColBERT Retrieval and Ensemble Response Scoring for Language Model Question Answering. arXiv preprint arXiv:2408.10808 (2024). doi:10.48550/arXiv.2408.10808
- [54] Oh, S., Kim, J., Park, J., Ko, S.W., Quek, T.Q.S., Kim, S.L.: Uncertainty-Aware Hybrid Inference with On-Device Small and Remote Large Language Models. In: *Proceedings of the 2025 IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, pp. 1–7. IEEE, Barcelona, Spain (2025). doi:10.1109/ICMLCN64995.2025.11140540
- [55] She, J., Zheng, W., Liu, Z., Wang, H., Xing, E.P., Yao, H., et al.: Token Level Routing Inference System for Edge Devices. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pp. 159–166. Association for Computational Linguistics, Vienna, Austria (2025). doi:10.18653/v1/2025.acl-demo.16
- [56] Cielen, D., De Bock, K.W., Florès, L.: Exploring Zero-Shot SLM Ensembles as an Alternative to LLMs for Sentiment Analysis. *Inf. Fusion* **126**, 103666 (2026). doi:10.1016/j.inffus.2025.103666
- [57] Li, M., Jiao, H., Zhou, T., Zhang, N., Peters, S., Lissitz, R.W.: Item Difficulty Modeling Using Fine-Tuned Small and Large Language Models. *Educ. Psychol. Meas.* **85**(6), 1065–1090 (2025). doi:10.1177/00131644251344973
- [58] Mehandru, N., Hall, A.K., Melnichenko, O., Dubinina, Y., Tsirulnikov, D., Bamman, D., et al.: BioAgents: Bridging the Gap in Bioinformatics Analysis with Multi-Agent Systems. *Sci. Rep.* **15**, 39036 (2025). doi:10.1038/s41598-025-25919-z
- [59] Amarasinghe, P.T., Nguyen, S., Sun, Y., Alahakoon, D.: Feedback-Integrated Prompt Optimiser for Problem Formulation. *Sci. Rep.* **15**, 43543 (2025). doi:10.1038/s41598-025-27495-8
- [60] Li, C., Chen, Q., Li, L., Wang, C., Li, Y., Chen, Z., et al.: Mixed Distillation Helps Smaller Language Model Better Reasoning. arXiv preprint arXiv:2312.10730 (2024). doi:10.48550/arXiv.2312.10730